

Jacen Li

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Education

University of Washington - Seattle

Expected June 2028

- B.S. - Electrical and Computer Engineering - Embedded Systems - GPA 3.85/4.0
- Courses: Comp Arch I & II, Hardware/Software Interface, Embedded Systems, Digital Design I & II, DSA

Skills

Languages: C, C++23, Python, Assembly (x86-64, ARM, RISC-V), Shell (Bash/Zsh)

Tools: Linux, Git/Github, perf, strace, GCC, GDB/objdump, Codex/Claude Code, Compiler Explorer

Experience

Arm

Summer 2026

Software Engineering Intern - System Performance Tooling

Austin, TX

Developed microbenchmarks for ASCT, the Arm System Characterization Tool

github.com/Arm/asct

- Evaluated 20+ core-to-core latency benchmarks across Intel, AMD, and Arm systems; compared timed operations, cache state, sampling methods, and prefetch sensitivity to guide ASCT benchmark design.
- Developed atomic-handoff (CAS) and directional pointer-chase benchmarks; mitigated prefetch interference and reordered pointer-chase setup to preserve the cache ownership being measured.
- Traced run to run variation to physical address dependent cache home selection; sampling multiple cache lines reduced std dev from 4.0->0.8 ns on Intel 8275CL and 4.7->0.1 ns on Graviton4, without shifting median latency.
- Fixed a thread-affinity race that pinned the control thread onto measured CPUs; used Linux scheduler and syscall traces to explain latency spikes and cut max observed scheduler delay from 4.0 ms to 78 us.
- Used firmware source, CMN captures, and PMU data to map physical addresses to cache-home nodes; built a homing-aware benchmark to vary the home for a fixed core pair and measure transfer latency.

National Institute of Standards and Technology

2023

Intern - Sensor Science Division

Gaithersburg, MD

- Deployed a 61-node I²C sensor network in the Tape Tunnel test chamber for dimensional metrology.
- Reduced thermal uncertainty by 38% through denser sampling; quantified error propagation w/ Monte Carlo.

Aeus Keyboards

2020 - 2025

Founder & Operator

jacen.li/keyboards.html

- Founded a custom keyboard business with over \$40K revenue; managed manufacturing and logistics.
- Designed keyboard PCBs in KiCad and cases for CNC machining; developed keyboard firmware in C using QMK.

Technical Projects

BLT - Better Latency Tester (STM32)

github.com/jtl06/BLT

- Built input-to-photon latency measurement device to benchmark and measure gaming PC responsiveness.
- Utilized timers, circular DMA, and ISRs for us-level resolution, quantified DMA ISR jitter: ± 8 us at 2.5 kHz.

Libdivide OSS contribution - Fast Division with ARM SVE

jacen.li/oss/libdivide

- Contributed ARM SVE/SVE2 SIMD paths for libdivide (merged), with tests, documentation, and Github CI fixes.
- Achieved ~5x higher 16bit and ~3x higher 64bit division throughput compared to NEON on AWS Graviton3.

SwiftFIX - SIMD FIX pre-parser for QuickFIX

github.com/jtl06/SwiftFIX

- Developed C++20 FIX preparser that scans, indexes, validates, and extracts into a zero-alloc offset table.
- Improved frame-splitting throughput from 146 MiB/s to 1.60 GiB/s scalar and 2.19 GiB/s with AVX2.

Nycticorax - Discord AI Agent

github.com/jtl06/Nycticorax

- Developed an AI agent with persistent memory, tool execution, provider recovery, and token/time budgets.
- Built an LLM evaluation harness for model A/B tests; scored success, token usage, quality and e2e latency.